

Certificate J: The Macbeath Rung $N = 9$

The cubic $x^3 - 3x + 1$, a cube root of unity, and the $\pi/9$ phase

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The “Dynamic Principle” program — the hodge-laboratory repository

Standalone theorem monograph · Monograph 18/20

1 Setting

The second cubic rung of the ladder is the level $N = 9$. The rung is named after Roderick Macbeath: the Macbeath curve of genus 7 — the Hurwitz surface with group $\mathrm{PSL}(2, 8)$ of order 504 — carries rotations of order 9, and the phase $\pi/9$ is its footprint in the program’s ladder. The $N = 9$ stand certifies the Fermat-level pipeline at $N = 9$: the Fermat curve X_9 has genus 28, the conductor census is nontrivial (two conductors, 3 and 9, already), and the minimal cubic $x^3 - 3x + 1$ is the simplest of the program’s three cubic floors. Certificate J secures: the census $28 = 1 + 27$, the exact cubic layer in $\mathbb{Z}[\zeta]/(\Phi_9)$, and the Cardano form whose arguments are primitive roots of unity.

Theorem 1 (J1 — the census of the level 3^2). *For the Fermat curve X_9 the conductor census of the eigencharacters (a, b) , $1 \leq a, b$, $a + b \leq 8$, is $\{3: 1, 9: 27\}$, and $\sum_d h_d = 28 = (9 - 1)(9 - 2)/2 = g$. Both schemes (the direct one and the Möbius one) agree.*

Proof. The conductor of a character is $d = 9/\mathrm{gcd}(9, a, b)$, so $d \in \{3, 9\}$. *Conductor 3:* one needs $\mathrm{gcd}(9, a, b) = 3$, i.e. $a, b \in \{3, 6\}$; the constraint $a + b \leq 8$ leaves the single pair $(3, 3)$ — the diagonal character; hence $h_3 = 1$. *Conductor 9:* $\mathrm{gcd}(9, a, b) = 1$; in total there are $c(9) = (9 - 1)(9 - 2)/2 = 28$ eigencharacters, so $h_9 = 28 - 1 = 27$. The Möbius scheme: $h_3 = c(3) - c(1) = 1 - 0 = 1$, $h_9 = c(9) - c(3) = 28 - 1 = 27$. Both schemes are integer identities. $\square$

Theorem 2 (J2 — the cubic layer of $b_{\mathrm{Ch}}(9)$). *The numbers $x_1 = \zeta + \zeta^{-1}$, $x_2 = \zeta^2 + \zeta^{-2}$, $x_4 = \zeta^4 + \zeta^{-4}$ with $\zeta = e^{2\pi i/9}$ have exact elementary symmetric polynomials*

$$s_1 = x_1 + x_2 + x_4 = 0, \quad s_2 = x_1x_2 + x_1x_4 + x_2x_4 = -3, \quad s_3 = x_1x_2x_4 = -1,$$

computed by pure integer arithmetic in $\mathbb{Z}[\zeta]/(\Phi_9)$; therefore x_1 is a root of the cubic $x^3 - 3x + 1$, and this is its exact minimal polynomial over $\mathbb{Q}$.

Proof. Sum. The exponents $\{\pm 1, \pm 2, \pm 4\} \bmod 9$ exhaust all the units of the cyclotomic field: $\{1, 2, 4, 5, 7, 8\}$. Hence $s_1 = \sum_{k \in (\mathbb{Z}/9)^\times} \zeta^k$ is the Ramanujan sum $c_9(1) = \mu(9) = 0$ (the level is divisible by 3^2 , where the Möbius function vanishes).

Pairwise products. $x_1x_2 = (\zeta + \zeta^8)(\zeta^2 + \zeta^7) = \zeta^3 + \zeta^8 + \zeta + \zeta^6$; $x_1x_4 = (\zeta + \zeta^8)(\zeta^4 + \zeta^5) = \zeta^5 + \zeta^6 + \zeta^3 + \zeta^4$; $x_2x_4 = (\zeta^2 + \zeta^7)(\zeta^4 + \zeta^5) = \zeta^6 + \zeta^7 + \zeta^2 + \zeta^3$. Here ζ^3 and ζ^6 are the primitive cubic roots: $\zeta^3 + \zeta^6 = \omega + \bar{\omega} = -1$. *Collecting:* $x_1x_2 = (-1) + (\zeta + \zeta^8)$, $x_1x_4 = (-1) + (\zeta^5 + \zeta^4)$, and $x_2x_4 = (-1) + (\zeta^7 + \zeta^2)$; *summing,* $s_2 = -3 + (\zeta + \zeta^2 + \zeta^4 + \zeta^5 + \zeta^7 + \zeta^8) = -3 + s_1 = -3$.

Product. $x_1x_2x_4 = (\zeta^3 + \zeta^8 + \zeta + \zeta^6)(\zeta^4 + \zeta^5) = \zeta^7 + \zeta^8 + \zeta^{12} + \zeta^{13} + \zeta^5 + \zeta^6 + \zeta^{10} + \zeta^{11} = \zeta^7 + \zeta^8 + \zeta^3 + \zeta^4 + \zeta^5 + \zeta^6 + \zeta + \zeta^2 = \sum_{k=1}^8 \zeta^k = -1$.

By Vieta’s formulas: $x^3 - s_1x^2 + s_2x - s_3 = x^3 - 3x + 1$. *Minimality:* $\mathrm{Gal}(\mathbb{Q}(\zeta_9)^+/\mathbb{Q}) \cong (\mathbb{Z}/9)^\times / \{\pm 1\} \cong C_3$ acts by multiplication by 2: the classes $\{1, 8\} \mapsto \{2, 7\} \mapsto \{4, 5\} \mapsto \{8, 1\}$ — a transitive cycle; hence $[\mathbb{Q}(x_1) : \mathbb{Q}] = 3$. *Fingerprint:* $x^3 - 3x + 1 \equiv x^3 + x + 1 \pmod{2}$ has no root in $\mathbb{F}_2$ ($0 \mapsto 1, 1 \mapsto 1$) — it is irreducible. $\square$

Theorem 3 (J3 — Cardano and the root of unity). *The cubic $x^3 - 3x + 1$ is already depressed ($p = -3$, $q = 1$); its Cardano discriminant is $\Delta = \frac{1}{4} - 1 = -\frac{3}{4} < 0$ — again the casus irreducibilis, and*

$$2 \cos \frac{2\pi}{9} = \sqrt[3]{\frac{-1 + \sqrt{-3}}{2}} + \sqrt[3]{\frac{-1 - \sqrt{-3}}{2}}, \quad b_{\text{Ch}}(9) = 1 - \frac{\sqrt[3]{\omega} + \sqrt[3]{\bar{\omega}}}{2},$$

where $\omega = \frac{-1 + \sqrt{-3}}{2} = e^{2\pi i/3}$ is a primitive cube root of unity. The principal cube roots are exactly $\sqrt[3]{\omega} = \zeta_9$ and $\sqrt[3]{\bar{\omega}} = \zeta_9^{-1}$: the Cardano arguments lie on the unit circle ($|\omega| = |\bar{\omega}| = 1$), and the product of the radicals is 1.

Proof. No depression is needed: the cubic $x^3 - 3x + 1$ already has the shape $y^3 + py + q$ with $p = -3$, $q = 1$. The discriminant is $\Delta = \frac{q^2}{4} + \frac{p^3}{27} = \frac{1}{4} - \frac{27}{27} = -\frac{3}{4} < 0$, $\sqrt{\Delta} = \frac{i\sqrt{3}}{2}$, and Cardano gives $y = \sqrt[3]{-\frac{1}{2} + \frac{i\sqrt{3}}{2}} + \sqrt[3]{-\frac{1}{2} - \frac{i\sqrt{3}}{2}} = \sqrt[3]{\omega} + \sqrt[3]{\bar{\omega}}$.

Principal branches. The argument of ω is $\frac{2\pi}{3} \in (-\pi, \pi]$; the principal cube root divides the argument by three: $\sqrt[3]{\omega} = e^{2\pi i/9} = \zeta_9$, and likewise $\sqrt[3]{\bar{\omega}} = e^{-2\pi i/9} = \zeta_9^{-1}$. Therefore $y = \zeta_9 + \zeta_9^{-1} = 2 \cos \frac{2\pi}{9}$ — the Cardano form reproduces the cyclotomic definition verbatim: at this rung the Cardano cube radicals are themselves ninth roots of unity. The branches are pinned by the modulus: $|\omega| = |\bar{\omega}| = 1$ — the only rung of the program for which both Cardano arguments lie on the unit circle; the product of the radicals $\sqrt[3]{\omega} \sqrt[3]{\bar{\omega}} = \sqrt[3]{\omega \bar{\omega}} = \sqrt[3]{1} = 1$ is an exact identity.

Isolation. The roots $x_1 \approx 1.53209$, $x_2 \approx 0.34730$, $x_3 \approx -1.87939$ are strictly ordered; at working precision (dps 60) the relative deviation of the Cardano form from $1 - \cos(2\pi/9)$ is zero within precision (protocol: $3.2 \cdot 10^{-36}$ at dps 35). $\square$

Remark 4. The diagonal character $(3, 3)$ of conductor 3 is the “hermit” of the census: the only character that does not lift to the full level 9. It is at the same time the trace of level 3: the self-similarity lemma (monograph 20/20) states $h_3(9) = h_3(3) = 1$ — every conductor carries a complete smaller stand. The order-9 rotations of the Macbeath curve give a geometric realization of the phase $\pi/9$: on the Hurwitz surface of genus 7 with $|\text{PSL}(2, 8)| = 504$ automorphisms the order 9 is maximal.

Protocol data: Certificate J — the stand numbers

Genus $g = 28$; census $\{3: 1, 9: 27\}$ (both schemes); minimal cubic $x^3 - 3x + 1$; Vieta $(s_1, s_2, s_3) = (0, -3, -1)$ exactly in $\mathbb{Z}[\zeta]/(\Phi_9)$; the cubic is irreducible over $\text{GF}(2)$; Cardano form $b_{\text{Ch}}(9) = 1 - \frac{\sqrt[3]{\omega} + \sqrt[3]{\bar{\omega}}}{2}$ with $\omega = \frac{-1 + \sqrt{-3}}{2}$; $\sqrt[3]{\omega} = \zeta_9$ (principal branch); the product of the radicals is 1; $b_{\text{Ch}}(9) = 0.233955556881021964797607349444583 \dots$; protocol relative errors: Cardano $3.2 \cdot 10^{-36}$, periods (spot sample) $2.3 \cdot 10^{-36}$ (dps 35); phase $\pi/9$.

Summary of results: summary

Certificate J accepted (5/5): the census $28 = 1 + 27$ by two schemes, the exact integer Vieta data $(0, -3, -1)$ in $\mathbb{Z}[\zeta]/(\Phi_9)$, irreducibility over $\text{GF}(2)$, the Cardano form with root-of-unity arguments ($|\omega| = 1$, the product of the radicals 1), and the phase $\pi/9$. The second cubic foundation of the ladder is established.

Verification: how to reproduce

`python3 laboratory.py` → “Certificates → J”; “Stands → N=9”; batch mode: a scenario with the bch item ($N = 9$); `--check-baseline` cross-checks $g = 28$, the census $\{3: 1, 9: 27\}$, the symmetric data $(0, -3, -1)$ and the errors against the reference.

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